Matrices, Vectors & Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

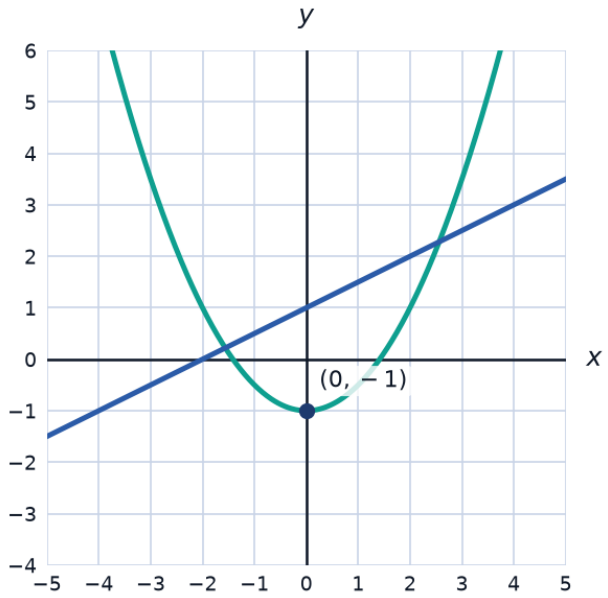
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, the complex number $z = x + yi$ satisfies the equation $z(2 - i) = 15 + 5i$ where $i^2 = -1$, and x and y are real numbers. Vector $\mathbf{v}$ is defined by $\mathbf{v} = [x; y]$. If matrix $A = [3, -1; 2, 4]$, what is the magnitude of the vector $A\mathbf{v}$? Show your complete algebraic work.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle k, 3 \rangle$, where $k > 0$. If the magnitude of the vector sum $\mathbf{u} + \mathbf{v}$ is $\sqrt{29}$, what is the value of k? Show your complete algebraic work.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle -3, 4 \rangle$. What is the vector $2\mathbf{u} + \mathbf{v}$? Show your complete algebraic work.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the complex plane, the point representing the complex number $z_1 = 3 + 4i$ is labeled A. The complex number z_2 is obtained by rotating z_1 counterclockwise by 90° about the origin, and its corresponding point is labeled B. The complex number z_3 is the complex conjugate of z_1, and its corresponding point is labeled C. What is the area of ΔABC? Show your complete algebraic work.

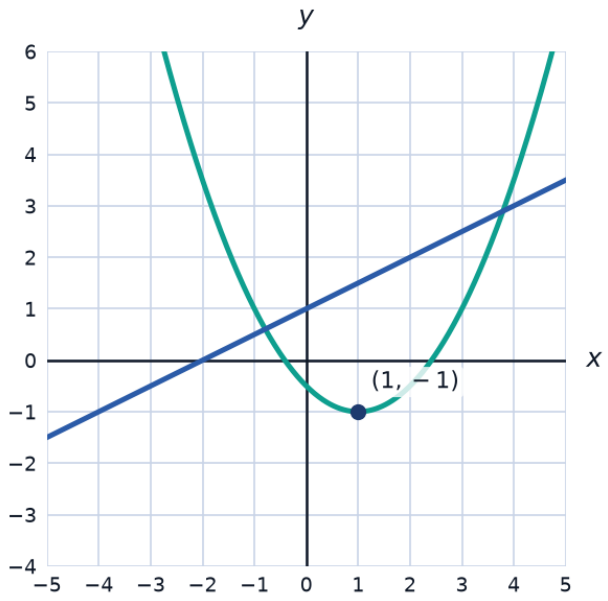
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the matrices $A = [2, -3; 1, 4]$ and $B = [-5, 1; 3, -2]$, what is the matrix $2A + B$? Show your complete algebraic work.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the product of $-4i$ and its complex conjugate.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the product of $6 + i$ and its complex conjugate.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -20, -21 \rangle$.

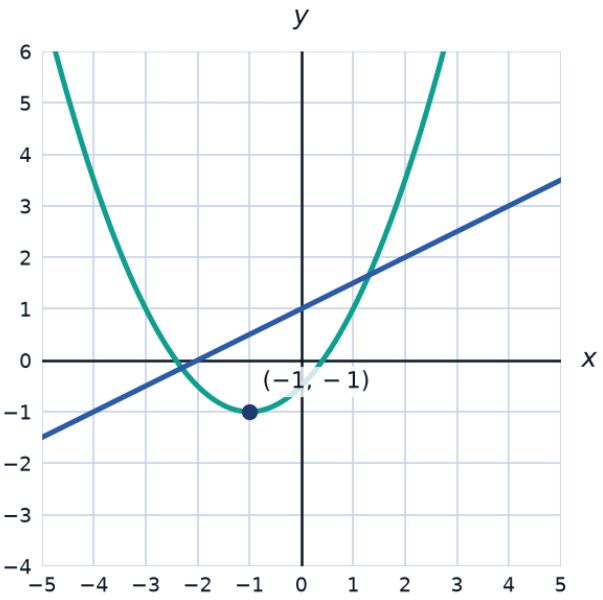
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -9, -40 \rangle$.

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10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -20, 21 \rangle$.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -24, -7 \rangle$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -9, 40 \rangle$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -5, 12 \rangle$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -7, 24 \rangle$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

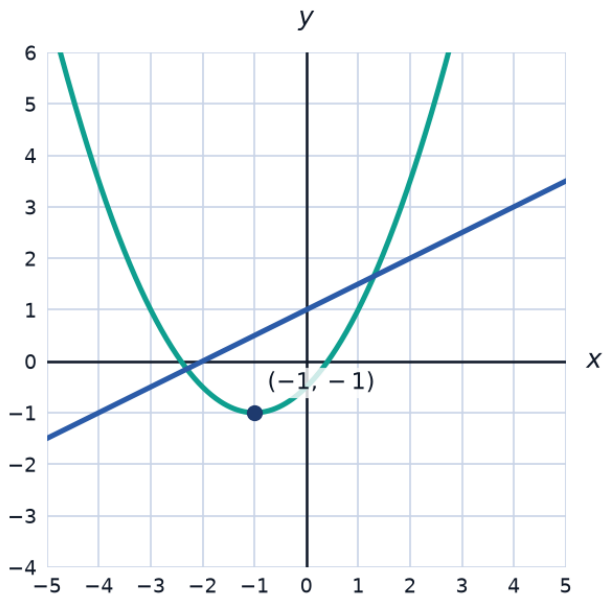
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Find the magnitude of the vector $\langle -8, 15 \rangle$.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the magnitude of the vector $\langle 15, 8 \rangle$.



17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

Find the magnitude of the vector $\langle 6, 8 \rangle$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

Find the magnitude of the vector $\langle -4, -3 \rangle$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

Find the magnitude of the vector $\langle 3, -4 \rangle$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

Find the product of $8 + 11i$ and its complex conjugate.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the product of the complex numbers $(4 - 3i)$ and $(2 + 5i)$, where $i^2 = -1$?

a. $23 + 14i$

b. $7 + 14i$

c. $23 - 14i$

d. $8 - 15i$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the simplified form of the quotient $\frac{5 - i}{2 + i}$?

a. $\frac{9 - 7i}{5}$

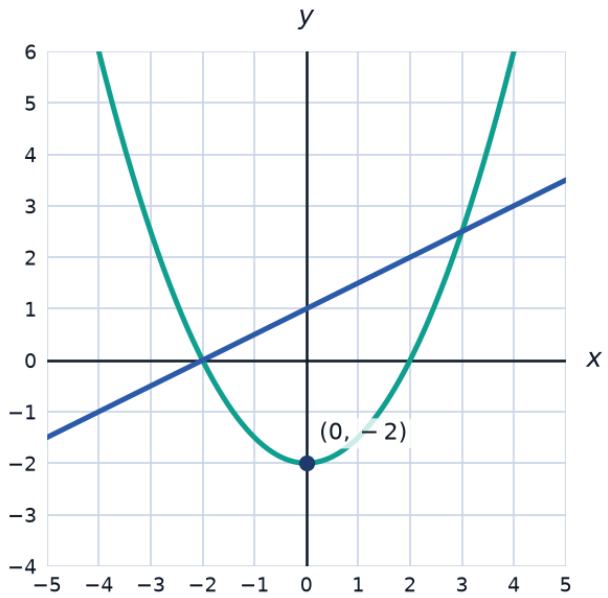
b. $\frac{11 - 7i}{5}$

c. $3 - i$

d. $\frac{9 + 3i}{5}$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the value of i^{47}, where $i = \sqrt{-1}$?



a. 1

b. -1

c. i

d. $-i$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the modulus (absolute value) of the complex number $z = -5 + 12i$?

a. 7

b. 13

c. 17

d. $\sqrt{119}$

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the distance in the complex plane between $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$?

a. $2\sqrt{13}$

b. $\sqrt{20}$

c. 10

d. 5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $z = 2 + 2i$, what is the argument θ of z in radians, where $0 \le \theta < 2\pi$?

a. $\frac{\pi}{6}$

b. $\frac{\pi}{4}$

c. $\frac{\pi}{3}$

d. $\frac{3\pi}{4}$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given matrices $A = [[3, -1], [2, 4]]$ and $B = [[1, 5], [-2, 3]]$, what is the matrix $2A - B$?

a. $[[5, -7], [6, 5]]$

b. $[[5, 3], [2, 5]]$

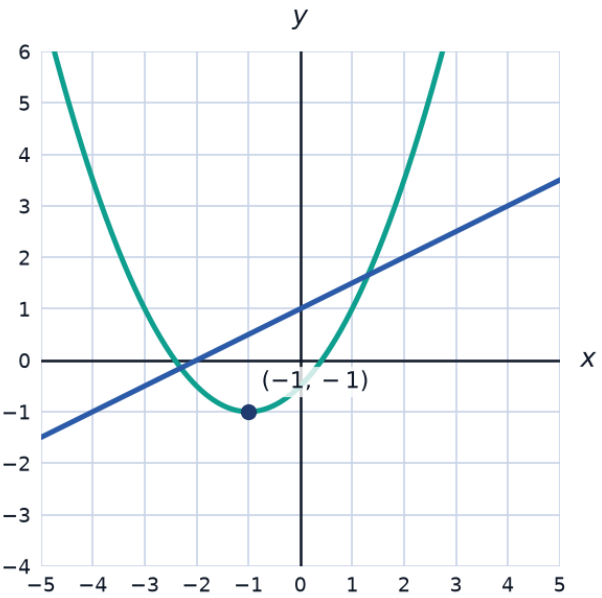
c. $[[7, -7], [6, 11]]$

d. $[[5, -7], [2, 5]]$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the determinant of the 2×2 matrix $M = [[4, 6], [-2, 3]]$?



a. 0

b. 12

c. 24

d. 36

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of k will the matrix $K = [[k, 6], [2, 3]]$ NOT have an inverse?

a. 1

b. 2

c. 3

d. 4

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$ are given. What are the components of $2\mathbf{u} + \mathbf{v}$?

a. $\langle 4, -3 \rangle$

b. $\langle 1, 1 \rangle$

c. $\langle 4, 3 \rangle$

d. $\langle -1, -3 \rangle$

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the dot product of the vectors $\mathbf{a} = \langle 5, -2 \rangle$ and $\mathbf{b} = \langle 4, 7 \rangle$?

a. 6

b. 26

c. -6

d. 34

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the magnitude of the vector $\mathbf{w} = \langle -9, 12 \rangle$?

a. 13

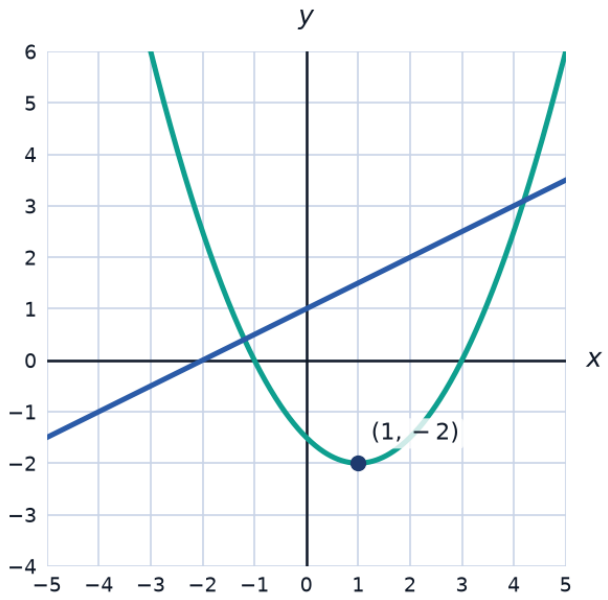
b. 15

c. 17

d. 21

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the product of $-4 + 5i$ and its complex conjugate.



a. 52

b. 8

c. 41

d. 35

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the product of $3 - 2i$ and its complex conjugate.

a. 13

b. 34

c. 1

d. 27

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35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the product of $4 - 3i$ and its complex conjugate.

a. 53

b. 11

c. 42

d. 25

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the product of $7 + 7i$ and its complex conjugate.

a. 111

b. 98

c. 123

d. 71

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the complex number system, for what real value of k is the expression $\frac{k+8i}{2-5i}$ a purely imaginary number? (Note: $i = \sqrt{-1}$)

a. -20

b. $-\frac{16}{5}$

c. $\frac{16}{5}$

d. 20

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

In the complex number system, where $i^2 = -1$, the expression $\frac{a+6i}{2-i}$ is equal to $4 + bi$ for some real numbers a and b. What is the value of $a + b$?

a. 5

b. 8

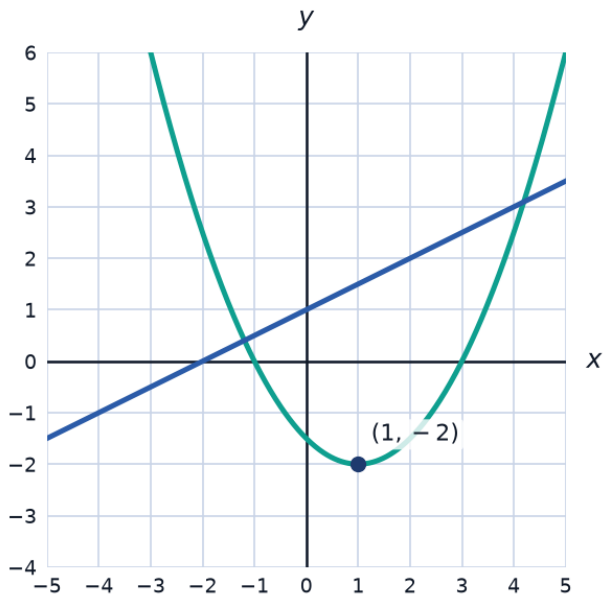
c. 13

d. 18

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39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

For the imaginary unit i, where $i^2 = -1$, which of the following complex numbers is equal to $(5 - 3i) - (2 - 8i)$?



a. $3 + 5i$

b. $3 - 11i$

c. $7 - 11i$

d. $3 - 5i$

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

If $A = (2, -11; 9, -3)$ and $B = (-14, -11; 11, -5)$, find the entry in row 1, column 2 of $A + B$.

a. -25

b. -21

c. -16

d. -22

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

If $M = (-3, -5; 6, -3)$, find the entry in row 2, column 1 of $2M$.

a. 16

b. 14

c. 12

d. 11

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

If $M = (-5, -1; 2, -4)$, find the entry in row 2, column 1 of $5M$.

a. 16

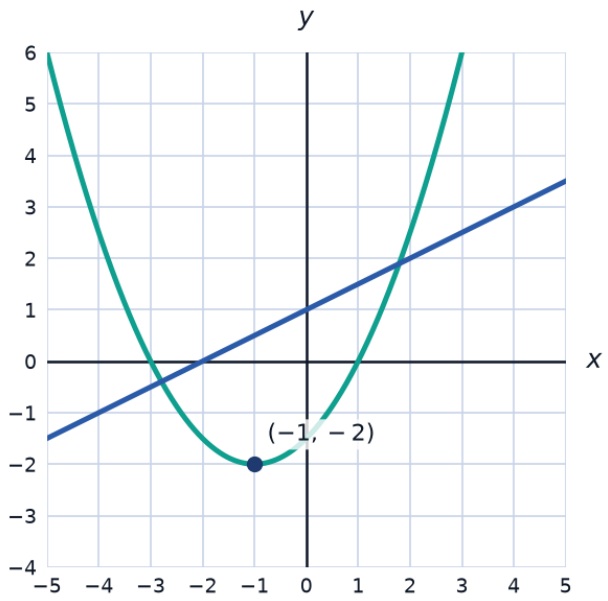
b. 15

c. 10

d. 8

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

If $M = (-5, -5; 4, -8)$, find the entry in row 2, column 1 of $4M$.



a. 16

b. 10

c. 18

d. 13

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

If $M = (-6, 2; 3, -7)$, find the entry in row 2, column 1 of $5M$.

a. 13

b. 10

c. 14

d. 15

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

If $M = (3, -7; 5, -1)$, find the entry in row 2, column 1 of $3M$.

a. 15

b. 8

c. 14

d. 17

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